R52: exact coordinate closure and larger source field

The complete nu adjunction stabilizes at index length 180, with least recovery denominator z^5(z+1)^9. An explicit degree 3 line norm supplies generic packet transport. Complete good fibers certify a degree 112 larger source field by monic specialization; all 3,810 stored factorizations are freshly checked.

Three constructive outputs and their next checks

Normal-chart closure

32 source-bound columns

Least denominator z^5(z+1)^9

Next: two completed local rings

Explicit image line

Full degree 3 norm identity

All three sign products

Next: Tate / QR / F2 descent

Larger marked-23 source

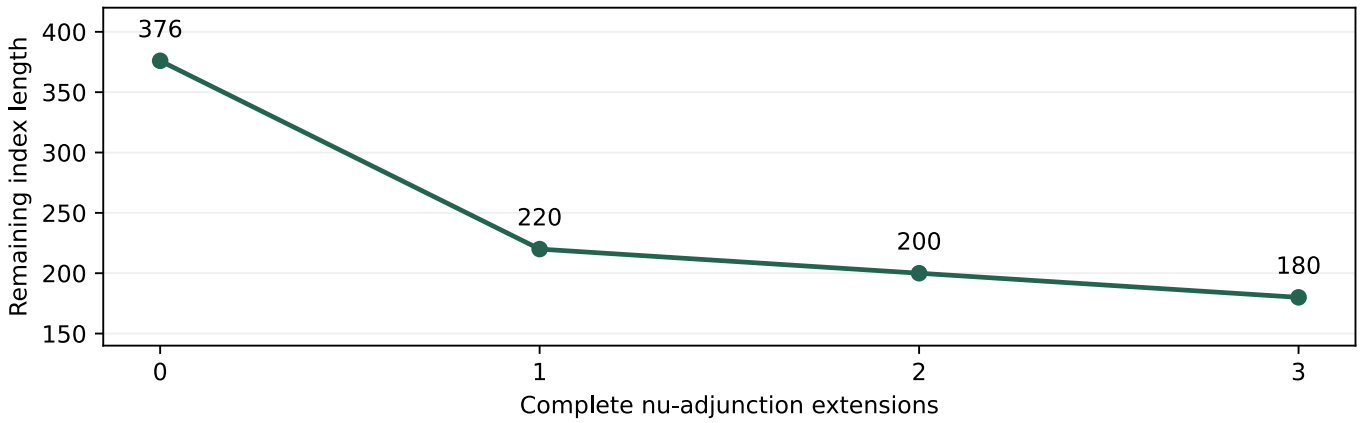
3,810 fresh factor certificates

Two-pattern degree 112 field

Next: normalization / 991 action

Complete coordinate algebra

Starting from S=F2[z][V, lambda] inside the retained rank 32 normal-chart B, all 32 actual nu-multiplication columns and their full source products pass. Both matrix products at every adjunction step and final stability pass.



The exact determinant is z^60(z+1)^120. The full 32x32 recovery product uses the least scalar z^5(z+1)^9; predecessor nonzero remainders prove minimality. A fresh process reconstructs every closure and source-action product.

Prime	Nonzero elementary exponents	Zeros	Length
z	1x4, 2x6, 3x6, 4x4, 5x2	10	60
z+1	1x4, 3x4, 4x4, 6x4, 7x4, 9x4	8	120

All 14 local levels retain both products. The complete types match the candidate (7,11) semigroup model (30 gaps, conductor 60), but actual completed local-ring identification still requires branch valuations and residue-field descent. The four coordinates recover the normal source on z(z+1)!=0, under the retained normal-RR bridge. Omitted fibers, infinity and proper polarization remain tasks.

Explicit line norm and the larger marked-23 field

Actual generic line transport

For f=y+lambda*x+nu and the degree 3 kernel {0, Q,-Q}, the full polynomial product is

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NormNumerator = alpha*Ynum*H^3 + beta*Xnum*H^4 + gamma*H^6
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Here H=x+t and alpha is the original line cubic evaluated at t. Consequently Y=(beta/alpha)X+gamma/alpha on alpha!=0. Changed intercept is rejected. The sign action preserves alpha, changes beta by A*alpha and gamma by b*alpha. The fresh checker reconstructs the actual translated line functions and Velu coordinates before checking the norm product and all 3 sign products.

Next certify alpha on the actual QR packet, normalize the marked point to Tate form, check all 3 target QR equations and gamma/F2 descent, including alternate charts and exceptional branches. The intended 11 primary action needs the actual integral/odd-adic correspondence module.

Larger source: two reducible patterns suffice

The retained marked23 base field K has degree 7 over F2(x). Its actual twisted QR polynomial is monic degree 16 and nu is linear with nonzero A^2Delta. Both fixed probes enumerate every coefficient-field element and every smooth linear base root with registered denominators and A*Delta nonzero.

Field degree	Good fibers	8+8	2+14
7	224	224	0
8	264	88	0
10	880	220	440
11	2442	242	484

The small range has 488 good fibers and no 2+14 pattern. Scaling gives 3,322 good fibers, 462 patterns 8+8, 924 patterns 2+14 and 2,024 odd-degree factors over F2048. Every 3,810 source specialization and each stored factor's irreducibility are freshly checked. Full factors and source coordinates are retained.

At a checked smooth base place, the local ring is a DVR and G's coefficients are integral. A monic global factor is integral there, so reduction preserves its degree. The 8+8 fiber allows degrees 0, 8, 16; the 2+14 fiber allows 0, 2, 14, 16. Only 0 or 16 survive, proving irreducibility and degree 112 over F2(x). A degree-dividing-10 base place rules out constant extension 7. An odd-degree 11 packet place then rules out a nontrivial constant field dividing 16, so the full constants are F2. The DVR/monic-specialization and place interpretations are external standard arguments in CM.md; the polynomial implication is formalized separately. Normalization of the 112 field and its selected 991 action are the next checks.

Receipts, formal scope and continuation

Terminal attempts: 17; accepted: 12; rejected: 5. All failed attempts remain in raw custody.

Attempt	Status	Native receipt
annihilator52-01	accepted	exit 0
closure-lean52-01	rejected	compile 1; audit not run
closure-lean52-02	accepted	compile 0; audit 0
closure52-01	rejected	exit 0; native errors
closure52-02	accepted	exit 0
fibers23-52-01	rejected	exit 0; marker absent
fibers23-52-02	accepted	exit 0
fibre-lean52-01	accepted	compile 0; audit 0
filtration52-01	rejected	exit 0; native errors
filtration52-02	accepted	exit 0
fresh52-01	accepted	exit 0
largefibers23-52-01	accepted	exit 0
normline52-01	accepted	exit 0
rect-lean52-01	accepted	compile 0; audit 0
reload52-01	rejected	exit 0; native errors
reload52-02	accepted	exit 0
semigroup52-01	accepted	exit 0

Nine declarations compile and pass the axiom audit with zero failing or trusted declarations. Closure52 proves range/stability/least-closure and unit-scalar implications plus the pattern intersection. FibreIrred52 proves polynomial irreducibility from the degree restrictions and the 7/10 constant divisor step. RectBridge52 covers the actual rectangular generator products. Actual native matrices, DVR specializations, source geometry and selected-factor actions are external; PROOF_MAP.md gives the precise boundary.

Native runs: seed 52, one Singular/FLINT thread,4 GiB virtual and 1200 s each. Lean4.34.1/mathlibd13f23b,-j1,-M4096,1200 s separately compile/audit, no VM cap. A native exit 0 or a late success marker never overrides error lines or a missing required marker. The first fiber loop was manually stopped after a documented continue-semantics bug; the corrected complete range was run once. Other rejections retain their source, logs and partial certificates.

Evidence: results.json, receipt-validation.json, CM.md, PROOF_MAP.md and the per-stage frozen protocols. ARTIFACTS.md records immutable byte custody and fresh download/repack; that is separate from the native fresh checker and external independent semantic review. Standard sources and exact references are recorded in CM.md. Existing sealed R27-R51 records are unchanged.

Continue with actual cusp valuations/completed subrings, normalized QR/Hecke transport,112 field normalization and 991 action. Preserve quotient saturation, proper polarized embeddings, node187 coupled divisors,132 section RR intersection and arbitrary reverse collectors, with Klein n3/samefield n5 controls.